

PATIENT & LABORATORY INFORMATION

- **Patient Name:** John/Jane Doe (Mock)
- **Date of Birth:** 05/12/1985
- **Date Collected:** 03/18/2026
- **Fasting Status:** Yes (12 Hours)

1. Complete Blood Count (CBC) with Differential

Evaluates overall health and detects a wide range of disorders, including anemia, infection, and leukemia.

Test Name	Result	Flag	Units	Reference Range
White Blood Cell (WBC)	6.8	Normal	x10 ³ /μL	3.8 - 10.8
Red Blood Cell (RBC)	4.90	Normal	x10 ⁶ /μL	4.20 - 5.80
Hemoglobin (Hb)	14.5	Normal	g/dL	13.2 - 17.1
Hematocrit (Hct)	43.5	Normal	%	38.5 - 50.0
Mean Corpuscular Vol (MCV)	88.0	Normal	fL	80.0 - 100.0
Mean Corpuscular Hgb (MCH)	29.5	Normal	pg	27.0 - 33.0
MCHC	33.5	Normal	g/dL	32.0 - 36.0
Platelet Count	245	Normal	x10 ³ /μL	140 - 400
Neutrophils (Absolute)	4.1	Normal	x10 ³ /μL	1.5 - 7.8
Lymphocytes (Absolute)	1.9	Normal	x10 ³ /μL	0.8 - 3.3
Monocytes (Absolute)	0.5	Normal	x10 ³ /μL	0.2 - 0.9
Eosinophils (Absolute)	0.2	Normal	x10 ³ /μL	0.0 - 0.5
Basophils (Absolute)	0.05	Normal	x10 ³ /μL	0.0 - 0.2

2. Comprehensive Metabolic Panel (CMP)

Provides information about the current status of your metabolism, including the health of your kidneys and liver, blood glucose, and electrolyte/acid-base balance.

Test Name	Result	Flag	Units	Reference Range
Glucose (Fasting)	112	HIGH	mg/dL	65 - 99
BUN (Blood Urea Nitrogen)	14	Normal	mg/dL	7 - 25
Creatinine	0.85	Normal	mg/dL	0.60 - 1.30
eGFR	>90	Normal	mL/min/1.73	>59

Test Name	Result	Flag	Units	Reference Range
BUN/Creatinine Ratio	16.5	Normal	n/a	9.0 - 20.0
Sodium	140	Normal	mmol/L	135 - 146
Potassium	4.2	Normal	mmol/L	3.5 - 5.3
Chloride	102	Normal	mmol/L	98 - 110
Carbon Dioxide (CO2)	26	Normal	mmol/L	20 - 32
Calcium	9.4	Normal	mg/dL	8.6 - 10.3
Total Protein	7.2	Normal	g/dL	6.1 - 8.1
Albumin	4.5	Normal	g/dL	3.6 - 5.1
Globulin	2.7	Normal	g/dL	1.9 - 3.7
Bilirubin, Total	0.6	Normal	mg/dL	0.2 - 1.2
Alkaline Phosphatase (ALP)	65	Normal	U/L	36 - 130
AST (SGOT)	22	Normal	U/L	10 - 40
ALT (SGPT)	28	Normal	U/L	9 - 46

3. Lipid Panel

Used to assess the risk of developing cardiovascular disease.

Test Name	Result	Flag	Units	Reference Range
Cholesterol, Total	225	HIGH	mg/dL	< 200
Triglycerides	140	Normal	mg/dL	< 150
HDL Cholesterol	48	Normal	mg/dL	> 40
LDL Cholesterol (Calc)	149	HIGH	mg/dL	< 100
Cholesterol/HDL Ratio	4.7	Normal	n/a	< 5.0
VLDL Cholesterol (Calc)	28	Normal	mg/dL	< 30

4. Thyroid & Iron Studies

Evaluates thyroid gland function and iron storage/transport in the body.

Test Name	Result	Flag	Units	Reference Range
TSH	2.45	Normal	mIU/L	0.40 - 4.50
Free T4	1.2	Normal	ng/dL	0.8 - 1.8
Iron, Total	85	Normal	µg/dL	50 - 170

Test Name	Result	Flag	Units	Reference Range
Total Iron Binding (TIBC)	320	Normal	µg/dL	250 - 450
Transferrin Saturation	26.5	Normal	%	15.0 - 50.0
Ferritin	65	Normal	ng/mL	30 - 400

5. Vitamins & Inflammatory Markers

Checks for common nutritional deficiencies and systemic inflammation.

Test Name	Result	Flag	Units	Reference Range
Vitamin D, 25-Hydroxy	18.5	LOW	ng/mL	30.0 - 100.0
Vitamin B12	450	Normal	pg/mL	200 - 1100
C-Reactive Protein (hs-CRP)	1.2	Normal	mg/L	< 3.0
Hemoglobin A1c (HbA1c)	5.8	HIGH	%	< 5.7

Clinical Laboratory Notes / Interpretation Summary:

- **Glucose & HbA1c:** Fasting glucose is mildly elevated at 112 mg/dL, and HbA1c is at 5.8%. These values fall into the "Prediabetes" range. Dietary modifications and exercise are usually recommended.
- **Lipid Panel:** Total Cholesterol (225) and LDL (149) are elevated indicating borderline-high risk for hyperlipidemia.
- **Vitamin D:** Value is 18.5 ng/mL, indicating a Vitamin D deficiency (optimal is >30 ng/mL). Supplementation is commonly advised by physicians for this level.
- **All other markers** (CBC, liver enzymes, kidney function, and thyroid) are within normal, healthy limits.